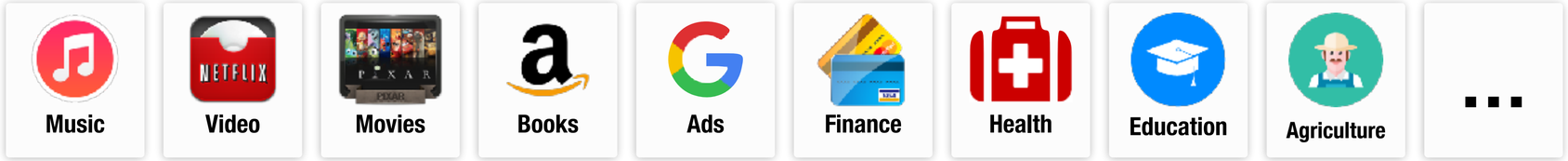
Software Dependability

Introduction to the course

Most businesses are being run on software with devastating risks due to software failures





This is problematic, but there’s something more…

Forecasts for Cybersecurity in 2026

Here are our forecasts for 2026 for the recent cyber security statistics:

- The global cyber security spending market will hit USD 240 billion in 2026, which is a 12.5% hike from 2025. New mandates like the AI Act and EU's NIS2 will force tighter governance, board-level accountability, and faster breach reporting.
- Annual global losses will hit around USD 10.5 trillion to USD 10.8 trillion at the rate of which cyber crime costs are going up. The US will be the single biggest market with an estimated total value of USD 81.61 billion.
- Rising loss frequencies also mean that the market will stay well-capitalized. We'll see an insurance shift and more signs of tightening will show up.
- Companies will deploy agentic SOCs to combat AI-powered threats. Shadow AI will continue to be a top risk but autonomous AI agents will handle up to 90% of routine triage.
- Double extortion attacks will keep growing in volumes, so recovering data alone won't be enough. Companies will focus on preventing leaks which will be their new priority. Post-quantum cryptography will move from theory to action planning as companies ready for the threats of "harvest now, decrypt later" attacks.

Such issues are not very far from us!

adnkronos

GUERRA UCRAINA RUSSIA IRAN CARBURANTI

Attacco hacker a TeamSystem, rubati Iban e dati personali dei clienti

L'accesso non autorizzato è del 24 agosto scorso. Oltre alle coordinate bancarie, sono stati trafugati anche e-mail e numeri di telefono. Ranieri Razzante all'Adnkronos: "Vigilate sui movimenti bancari di questi giorni"



Polizia Postale al lavoro - Fotogramma

Who am I?



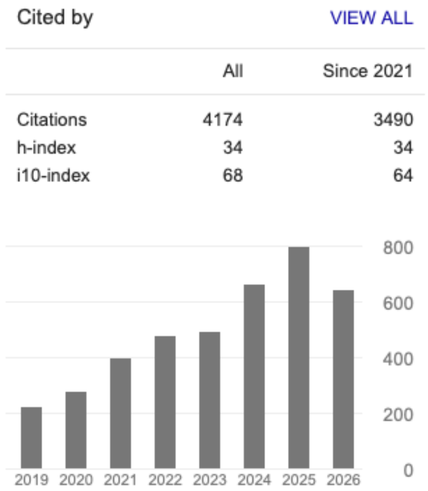
Who I am?

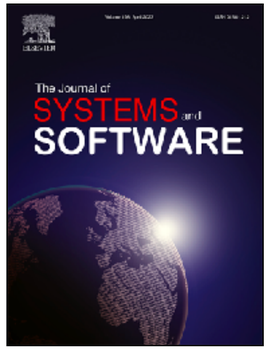
- Lecture of Software Dependability (M.Sc. in Computer Science)
- Lecturer of Software Engineering for Secure Cloud Systems (M.Sc. in IT Security and Cloud Technologies)
- Lecturer of Object Oriented Programming (B.Sc. in Computer Science)

empirical software engineering
software maintenance and evolution
software testing
sustainable software engineering

sesa lab
SOFTWARE ENGINEERING
SALERNO

machine learning
search-based algorithms
mining of software repositories
quantum computing

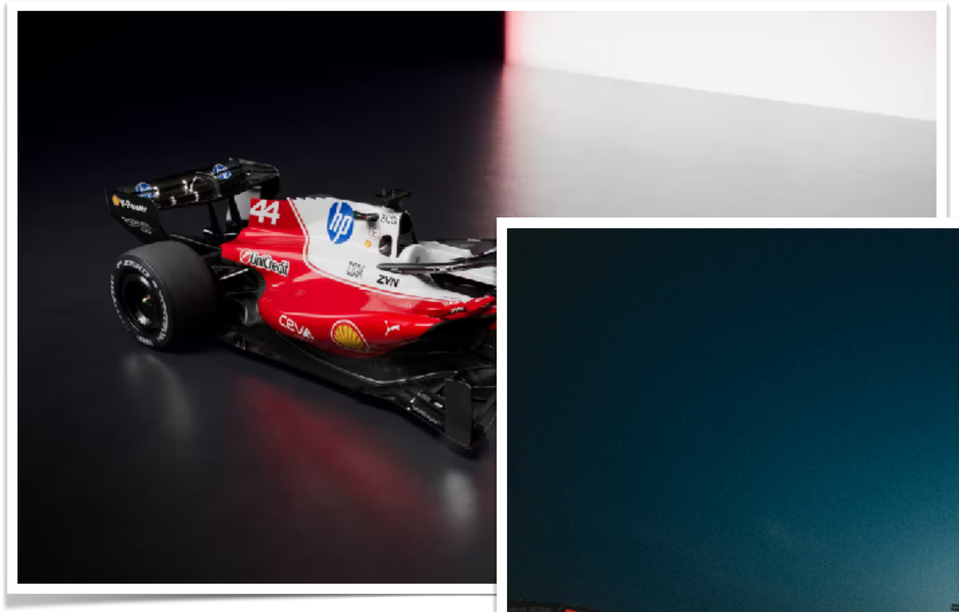




ACM Transactions on Software Engineering and Methodology

IEEE Software

Who I am?







sesa lab
SOFTWARE ENGINEERING SALERNO

3 Full Professors, 2 Associate Professors, 1 Assistant Professor, 8 Ph.D. Students

Education

- Software Engineering (B.Sc.)
- Object Oriented Programming (B.Sc.)
- Programming and Data Structures (B.Sc)
- Fundamentals of Artificial Intelligence (B.Sc.)
- Software Engineering Advanced Techniques (M.Sc.)
- Software Metrics, Analytics, and Quality (M.Sc.)
- Software Project Management (M.Sc.)
- **Software Dependability (M.Sc.)**
- Sustainable Software Engineering (M.Sc.)
- Software Engineering for A.I. (M.Sc.)
- Computer Science Education (M.Sc.)

Research

- Software Maintenance and Evolution
- Software Testing
- Performance Testing
- Blockchain Software Engineering
- Quantum Software Engineering
- Requirement Engineering & Artificial Intelligence
- Social Software Engineering
- Software Project Management
- Diversity & Inclusion of Software Systems
- Fairness of Software Systems

JADS Jheronimus Academy of Data Science
TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY
VUB VRIJE UNIVERSITEIT BRUSSEL
UNIVERSITY OF HAWAI'I
UNIVERSITY OF OULU
THE UNIVERSITY OF AUCKLAND Te Whare Wānanga o Tāmaki Makaurau NEW ZEALAND
WILLIAM & MARY CHARTERED 1693
POLYTECHNIQUE MONTRÉAL UNIVERSITÉ D'INGÉNIERIE

https://sesalabunisa.github.io/

The Need for Software Dependability

The dependability of a system reflects the extent of the user's confidence when operating and interacting with the system. A system can be considered dependable if (i) it operates without interruption, (ii) delivers the services that the stakeholders expect, (iii) does not adversely affect the environment, and (iv) does not damage its data or the system itself.

M1: Fundamentals of Software Dependability

- Intro to Software Dependability
- Dependability Requirements, Faults, and Treatments
- Software Fault Avoidance in Specification and Implementation
- Software Fault Tolerance and Assessment
- IoT Architectures
- DevOps and Reliable Programmable Infrastructure

M2: Software Testing

- Testing of Cyber-Physical Systems
- Intro to Software Testing and Test Case Selection
- Test Case Assessment and Test-Driven Development
- Performance Testing

M3: Detection of Software Vulnerabilities

- Introduction to Software Vulnerabilities
- Static and Dynamic Techniques to Detect and Exploit Software Vulnerabilities

M4: Software Dependability Lab







Administrative Information - General

lectures	Thu: 11:00 - 13:00 Fri: 11:00 - 14:00
e-learning platform	https://elearning.informatica.unisa.it/el-platform/course/view.php?id=1266
email	ddinucci@unisa.it
office hours	Wed 14:00 - 16:00 and Thu 14:00 - 16:00 Please contact me beforehand.

Administrative Information - Calendar

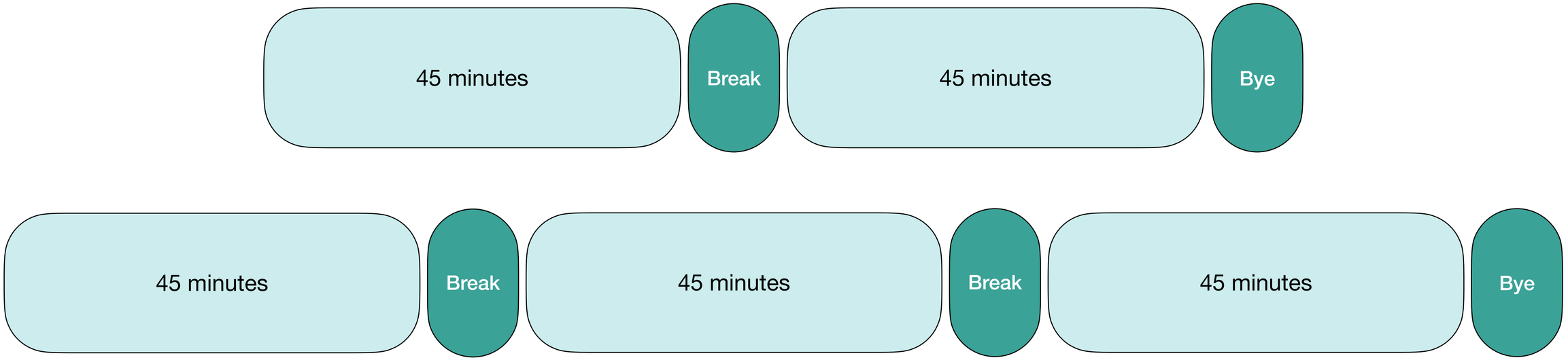
	Thursday (11:00-13:00) P6	Friday (11:00-14:00) P6
Week 1 (21 Sep - 25 Sep)	Intro to the course Intro to Software Dependability	Dependability Requirements, Faults, and Treatments
Week 2 (28 Sep - 2 Oct)	Software Fault Tolerance and Assessment	Software Fault Avoidance in Specification and Implementation
Week 3 (5 Oct - 9 Oct)	IoT Architectures	DevOps and Dependable Programmable Infrastructure
Week 4 (12 Oct - 16 Oct)		
Week 5 (19 Oct - 23 Oct)	Testing of Cyber-physical Systems	Intro to Software Testing and Test Case Selection
Week 6 (26 Oct - 30 Oct)	Test Case Assessment and Test-Driven Development	Performance Testing Introduction to Software Vulnerabilities
Week 7 2 Nov - 6 Nov)	Software Vulnerability Atelier	Dynamic Techniques to Detect and Exploit Software Vulnerabilities

	Thursday (11:00-13:00) P6	Friday (11:00-14:00) P6
Week 8 (9 Nov - 13 Nov)	JML and OpenJML	Docker, Docker Compose, and Podman
Week 9 (16 Nov - 20 Nov)	Docker Swarm and Kubernetes GitHub Workflows	JUnit and Mockito
Week 10 (23 Nov - 27 Nov)	JaCoCo, CodeCov, and PiTest	Java Microbenchmark Harness, JU2JMH, and AMBER
Week 11 (30 Nov- 4 Dec)	Code security analysis integrated in GitHub	GitGuardian, Snyk, SonarQube Cloud
Week 12 (7 Dec - 11 Dec)	Open Lab	Open Lab

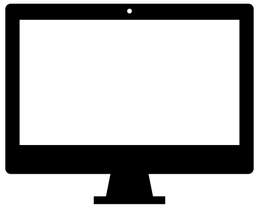
Administrative Information - Lecture Structure

The ability to stay focused on contents that are conveyed to us and that require constant mental effort, focused on a goal, is known as *sustained attention*. To keep such a sustained attention constantly high, it is recommended to interleave 45 minutes of lecture with 15 minutes of break.

Hence, whenever possible, the lecture will be structure along those lines:



Administrative Information - Exam Policy and Dates



Project

This grade counts for 50% of the final grade.
You **must** submit the project on the e-learning platform one week before the oral exam.



Oral Exam

This grade counts for 50% of the final grade.
It covers all module courses. By law, you must take exams in person.

January 18th, 9 AM, F5 Room
February 1st, 9 AM, F5 Room
February 15th, 9 AM, F5 Room

Before the Exam

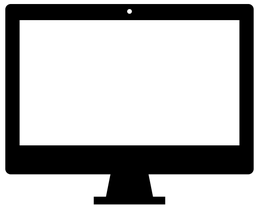
Complete the project and deliver the required documentation one week before the oral exam

→

Oral Exam

During the oral exam I will ask you questions about the project

Administrative Information - Project



Project

This grade counts for 50% of the final grade.
You **must** submit the project on the e-learning platform one week before the oral exam.

You have **three options**.

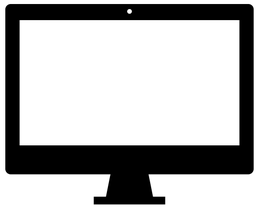
Option #1 - Dependability analysis of a Java open-source project. You will learn which projects are worth analyzing during the lab sessions. However, please double-check the project's suitability with me BEFORE starting! Some open-source projects might be too trivial/complicated to touch/test —> be smart and ask first!
For a part of the project, you will need a web app; therefore, you will need to wrap your Java app in a web app to use it.

Option #2 - Dependability analysis of your Java web project. If you developed a complex enough web project, you could apply the methods studied in the course directly to that.

Option #3 - Have a project in combination with other courses. If you are developing or extending a Java web project for other courses, let us discuss it.

The projects can be conducted in teams composed of **2/3 people or alone**.

Administrative Information - Project



Project

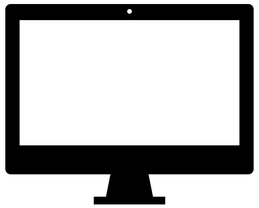
This grade counts for 50% of the final grade.
You **must** submit the project on the e-learning platform one week before the oral exam.

Options #1, #2, and #3.
The goal is to measure the dependability of these systems through empirical investigations of the source code or the related change history, report on these aspects, and possibly provide developers with insights on how to structure the code or the development process better.
Your report must cover the course concepts; in particular, you must apply the tools shown in the lab sessions to the selected software system. For each applied tool, you will receive points that will contribute to the final project grade.

Options #1.
Depending on the projects, you may want to open pull requests and contribute to open source.

Option #3.
The final grade will be independent of the other evaluations; particularly, it will focus on the attention to software dependability in the developed project.

Administrative Information - Project



Project

This grade counts for 50% of the final grade.
You **must** submit the project on the e-learning platform one week before the oral exam.

Each group must create a GitHub repository for all documents and code. If there is no repository at the end of the semester or the repository is incomplete, **this part of the exam will be considered failed.**

Each group member must contribute significantly to the project.
We will analyze the git logs to verify your contribution: no commits means no contribution.
As such, **using Git appropriately is key.** Forget USB pens, Google Drive, or Dropbox folders!

All documentation must be part of the GitHub repo, stored in **Markdown** files.

Administrative Information - What are my expectation?

Study lesson by lesson and apply theoretical concepts as they are introduced.
Come to the lab sessions and get your project done step by step.

Do you have any doubt?

Contact me via mail

Talk with me at the end of the lecture

Schedule a meeting with me (in presence or online)


Break